# GRNScope

## 1. User Authentication and Account Management

GRNScope provides a lightweight account system that enables persistent storage of analyses and long-term access to results. The platform supports three authentication-related functions.

**1.1 Registration.** A visitor may create an account by providing an email address and a password of at least eight characters. The email serves as the unique login identifier. Upon successful registration, the user is automatically authenticated and directed to their project dashboard. Duplicate email addresses are rejected with an appropriate error message.

**1.2 Login and Session Management.** A registered user authenticates by supplying their email and password. Upon successful verification, the system issues a session token valid for 24 hours. All subsequent interactions with protected resources are authorized via this token. When the token expires, the user must re-authenticate. There is no "remember me" or refresh-token mechanism in the initial release.

**1.3 Anonymous Access.** The platform permits limited use without an account. An unauthenticated visitor may browse the algorithm directory and the landing page statistics. If the platform later supports anonymous job submission, results from anonymous sessions expire and are permanently deleted after 72 hours.

## 2. Project Management

Projects are the top-level organizational unit. Every dataset, job, and result set belongs to exactly one project, and every project belongs to exactly one user.

**2.1 Create Project.** An authenticated user may create a new project by specifying a name and an optional free-text description. There is no limit on the number of projects a user may own.

**2.2 List Projects.** The project dashboard displays all projects belonging to the authenticated user, sorted by creation date (most recent first). Each entry shows the project name, creation timestamp, the number of datasets uploaded, and the number of analysis jobs submitted.

**2.3 View Project Detail.** Selecting a project opens a detail view that serves as the hub for all operations within that project: uploading data, launching analyses, monitoring job progress, and accessing results.

**2.4 Delete Project.** A user may permanently delete a project. Deletion cascades to all associated datasets, jobs, task records, and stored files (both uploaded data and computed results). This action is irreversible and requires explicit user confirmation.

# 3. Data Upload and Validation

The upload function ingests the user's experimental data and prepares it for downstream analysis. GRNScope accepts single-cell RNA sequencing expression matrices as its primary input.

**3.1 Expression Matrix Upload.** The user uploads a gene expression matrix in CSV format via a drag-and-drop interface or file browser dialog. The expected format is a matrix in which rows represent genes, columns represent individual cells, and values are normalized RNA expression counts. The first row contains cell identifiers and the first column contains gene names. Upon receipt, the system performs the following validation steps: it verifies that the file is a well-formed CSV, that it does not exceed the maximum permitted file size (500 MB), that the header row and first column are parseable as identifiers, and that the interior values are numeric. The system extracts and records the number of genes, the number of cells, and the complete list of gene names. These metadata are stored for display in the user interface and for downstream use by the preprocessing and visualization subsystems.

**3.2 Pseudotime Upload (Optional).** Certain GRN inference algorithms require a pseudotemporal ordering of cells — a one-dimensional trajectory that approximates the biological progression (e.g., differentiation) captured in the dataset. The user may optionally upload a pseudotime file, which is a single-column CSV containing one floating-point value per cell, ordered consistently with the columns of the expression matrix. The system validates that the number of pseudotime values matches the number of cells in the expression matrix.

**3.3 Automatic Pseudotime Computation (Planned).** When the user requires pseudotime but does not have a precomputed ordering, the platform offers the option to compute pseudotime automatically using the Slingshot algorithm. When this option is enabled, the user specifies the starting cell cluster and the expected number of trajectories. The computation runs as a preprocessing step before algorithm execution.

**3.4 Sample Dataset Download.** For new users or demonstration purposes, the landing page and the upload interface provide a downloadable sample dataset with a known structure. This allows users to explore the full analysis workflow without preparing their own data.

# 4. Data Preprocessing

After upload and before algorithm execution, the expression matrix undergoes a configurable preprocessing pipeline. Preprocessing is a discrete step in the analysis wizard, and the exact parameters chosen are recorded alongside the results for reproducibility.

**4.1 Gene Filtering by Variability.** The user specifies the number of top most-variable genes to retain (default: 2,000). Gene variability is computed as the variance of expression values across all cells. This filtering step serves two purposes: it reduces the computational cost of inference algorithms (which scale superlinearly with gene count), and it focuses the analysis on genes with sufficient expression variation to support regulatory inference. The BEELINE benchmarking study found that filtering to 500 or 1,000 genes provided a practical tradeoff between coverage and computational feasibility.

**4.2 Transcription Factor Inclusion Override.** Independent of the variability filter, the user may enable an option to retain all known transcription factors regardless of their variability rank. This is motivated by the BEELINE paper's finding that including all significantly varying transcription factors produced statistically significant improvements in early precision ratio. When enabled, the system consults an internal reference list of known TF gene names and ensures they are present in the filtered gene set even if they fall outside the top-N variability cutoff.

**4.3 Normalization.** A toggle (enabled by default) applies library-size normalization to the expression matrix, scaling each cell's total expression to a common value. This corrects for technical variation in sequencing depth across cells.

**4.4 Log-Transformation.** A toggle (enabled by default) applies a $\log_2(x + 1)$ transformation to all expression values after normalization. This compresses the dynamic range of expression counts, which is a standard preprocessing step for scRNA-seq data and is expected by most inference algorithms.

**4.5 Preprocessing Summary.** Before the user proceeds to algorithm selection, the interface displays a summary of the effect of their chosen parameters — for example, "Your 18,000 genes × 12,400 cells matrix will be filtered to 2,000 genes." This feedback helps users make informed parameter choices.

# 5. Algorithm Directory

The algorithm directory is a publicly accessible catalog of all GRN inference methods supported by the platform. It serves both as a reference resource and as the foundation for the algorithm selection step of the analysis wizard.

**5.1 Browse Algorithms.** The directory presents all active algorithms in a card-based grid layout. No authentication is required to view the directory. Each card displays the algorithm name, a one-sentence description of its computational approach, its methodology category (e.g., Machine Learning, Correlation, Information Theory, ODE-based, Granger Causality, Boolean), property badges indicating whether the algorithm outputs directed edges, signed edges, or requires pseudotime input, the publication year and journal of the original method paper, and the estimated runtime per standard dataset size.

**5.2 Filter Algorithms.** A persistent sidebar provides multi-select filters for methodology category, pseudotime requirement, edge directionality, and edge signedness. Filters are applied with AND logic within a category and OR logic across values within a single filter. The card grid updates immediately as filters are toggled.

**5.3 View Algorithm Detail.** Clicking an algorithm card opens a detailed view (either a modal overlay or a dedicated page) that presents the full description of the method, its known strengths and limitations as characterized by the BEELINE benchmarking study, recommended use cases, the version of the Docker container currently deployed on the platform, and links to the original publication and source code repository.

**5.4 Algorithm Contribution (Community).** Authenticated users may propose new algorithms for inclusion via a submission form. The form collects the algorithm name, a description, a link to the source code, and optionally a pre-built Docker image tag. Submissions enter a review queue visible to platform administrators.

# 6. Algorithm Selection and Job Configuration

This function constitutes the third step of the analysis wizard and determines which inference methods will be executed on the user's preprocessed data.

**6.1 Individual Algorithm Selection.** The user selects one or more algorithms from a card-based interface. Cards are organized into two groups: methods that do not require pseudotime, and methods that do. Each card displays the same information as in the algorithm directory, plus a checkbox for selection and an estimated runtime badge. If the uploaded dataset does not include pseudotime data (and automatic computation was not enabled), the pseudotime-dependent cards are visually disabled and display an explanatory message.

**6.2 Preset Selections.** A "Recommended" preset button selects a curated combination of algorithms representing each methodology category. This lowers the barrier to entry for users who are unfamiliar with the relative merits of individual methods. The recommended set is informed by the BEELINE paper's findings: it includes PIDC, GENIE3, and GRNBoost2 as the most consistently strong performers.

**6.3 Select All.** A convenience button selects all algorithms compatible with the current dataset (i.e., all pseudotime-independent methods, plus pseudotime-dependent methods if pseudotime is available).

**6.4 Ensemble Analysis Toggle.** A toggle at the bottom of the selection interface enables ensemble (consensus) analysis. When enabled, the platform will compute a consensus network aggregating the predictions of all selected methods, in addition to storing each method's individual output. This toggle is enabled by default when two or more algorithms are selected.

**6.5 Job Review and Submission.** The final wizard step presents a read-only summary of all user choices: dataset name and dimensions (after preprocessing), preprocessing parameters, selected algorithms, and estimated total runtime. The user confirms and submits the analysis. Upon submission, the system acknowledges receipt and the user may navigate away from the page; results will be available asynchronously.

## 7. Job Execution and Monitoring

Once submitted, an analysis job enters an asynchronous execution pipeline. The monitoring function provides the user with real-time visibility into the progress of their analysis.

**7.1 Parallel Algorithm Execution.** Each selected algorithm executes independently and in parallel within its own isolated container environment. This architectural choice ensures that a failure in one algorithm does not affect the others, and that fast algorithms deliver results without waiting for slow ones to complete.

**7.2 Per-Algorithm Status Tracking.** The job monitoring interface displays the status of each algorithm as an independent item. Possible statuses are Queued (waiting for a worker), Running (actively executing), Completed (finished successfully), and Failed (terminated with an error). For running tasks, the elapsed time since execution began is displayed.

**7.3 Overall Job Status.** The job itself has an aggregate status derived from its constituent tasks: Queued (no tasks have started), Running (at least one task is executing), Completed (all tasks finished successfully), Partially Completed (at least one task succeeded and at least one failed), and Failed (all tasks failed).

**7.4 Progressive Results Access.** The user does not need to wait for all algorithms to finish before beginning to explore results. As soon as any individual algorithm completes, its results become available for viewing. This is particularly valuable when a job combines fast correlation-based methods (completing in minutes) with slow machine-learning methods (requiring tens of minutes or hours).

**7.5 Failure Reporting.** When an algorithm fails, the monitoring interface displays a truncated error message extracted from the algorithm container's standard error output. The user may expand the message to view the full error text, which aids in diagnosing input format issues or algorithm-specific failures.

**7.6 Polling Mechanism.** The monitoring interface automatically refreshes job status at regular intervals (every five seconds) while any task remains in a non-terminal state. Polling ceases once all tasks have reached a terminal state, reducing unnecessary network traffic.

## 8. Consensus Network Computation

The consensus function is the central analytical capability of GRNScope. It synthesizes the outputs of multiple GRN inference algorithms into a single, higher-confidence network by retaining only regulatory edges that are independently predicted by multiple methods.

**8.1 Multi-Algorithm Aggregation.** For each pair of genes (Source, Target), the system determines how many of the selected algorithms predicted a regulatory edge between them. Each algorithm's output is a ranked list of edges; only the top N edges per algorithm are considered (see 8.3). An edge is included in the consensus network if and only if the number of algorithms supporting it meets or exceeds the user-defined consensus threshold (see 8.2).

**8.2 Consensus Threshold.** The user controls a threshold parameter — an integer ranging from 1 (include any edge predicted by at least one algorithm) to the total number of selected algorithms (include only edges predicted by every algorithm). The threshold is adjustable in real time via a slider in the results workspace. Changing the threshold instantly recomputes and re-renders the consensus network without any server round-trip, as the computation runs entirely in the browser.

**8.3 Top-N Edge Filtering.** Before consensus aggregation, each algorithm's output is truncated to its top N highest-ranked edges (default: 5,000). This parameter is also user-adjustable. Its purpose is to exclude low-confidence predictions that an algorithm ranks near the bottom of its output, which would otherwise inflate the consensus count for spurious edges.

**8.4 Consensus Scoring.** Each edge that survives the threshold filter receives a consensus score computed as the mean of the normalized confidence scores from all contributing algorithms. Edges are then ranked by consensus score in descending order, producing a unified ranked edge list.

**8.5 Design Rationale.** The consensus approach is motivated by the observation, documented in the BEELINE study, that different algorithms exhibit complementary strengths: an edge correctly predicted by multiple methodologically diverse algorithms is more likely to represent a genuine regulatory interaction. However, the BEELINE paper also found that ensembles do not always outperform the single best method, which is why GRNScope preserves individual algorithm outputs alongside the consensus view and allows the user to toggle freely between them.

# 9. Network Visualization

The network visualization function renders the inferred gene regulatory network as an interactive, explorable graph. It is the primary interface through which users interpret their results.

**9.1 Graph Rendering.** The network is displayed as a node-link diagram on an interactive canvas occupying the majority of the results workspace viewport. Nodes represent genes. Directed edges represent predicted regulatory relationships. The graph is rendered using a force-directed layout algorithm by default, which positions densely connected gene clusters together and separates sparsely connected regions.

**9.2 Visual Encoding.** Node appearance encodes gene type: transcription factor nodes are rendered as diamonds in a distinct color (teal), while target gene nodes are rendered as circles in a neutral color (slate gray). Node size is proportional to the node's degree (total number of edges). Edge width is proportional to the consensus score (or the individual algorithm's confidence score when viewing a single method's output). Edge color encodes the number of supporting algorithms, ranging from light gray (one algorithm) to saturated teal (all algorithms).

When viewing the output of a single algorithm that produces signed edges, edge color instead distinguishes activating relationships (blue) from repressive relationships (red).

**9.3 Interactive Exploration.** The canvas supports panning (click-and-drag on empty space), zooming (mouse wheel or pinch gesture), and node selection (click). The user may also drag individual nodes to manually adjust the layout, with the node remaining at its new position.

**9.4 Method Switching.** A dropdown control allows the user to switch between viewing the consensus network and the output of any individual selected algorithm. Switching the view updates the graph immediately, recomputing visible edges and re-applying the layout.

**9.5 Layout Options.** The user may switch between multiple graph layout algorithms: force-directed (the default, emphasizing cluster structure), hierarchical (arranging regulators above their targets), concentric (placing high-degree nodes at the center), and circular (arranging all nodes on a ring). Layout changes animate smoothly.

**9.6 Node Inspection.** Clicking a node selects it and populates the Node Inspector panel (in the right sidebar) with detailed information: the gene name, whether it is classified as a transcription factor, its in-degree (number of predicted regulators), its out-degree (number of predicted targets), a ranked list of its top regulators (by edge confidence), and a ranked list of its top target genes. This allows a biologist to quickly assess the predicted regulatory role of any gene of interest.

**9.7 Edge Tooltip.** Hovering over an edge displays a tooltip showing the source gene, target gene, consensus score (or individual score), consensus rank, and the names of all algorithms that predicted this edge.

**9.8 Gene Search.** A search bar with autocomplete allows the user to locate a specific gene by name. Selecting a gene from the search results centers the canvas on that node and selects it, triggering the Node Inspector update.

**9.9 Sub-Network Isolation.** For any selected node, the user may click an "Isolate Sub-network" button. This hides all nodes and edges not directly connected to the selected gene, re-runs the layout on the visible subgraph for optimal readability, and displays a "Reset View" button to restore the full network. This function is essential for exploring the local regulatory neighborhood of a gene of interest within a large, dense network.

# 10. Edge Analysis Table

The edge analysis table provides a tabular, sortable, and filterable view of the complete ranked edge list. It serves users who prefer quantitative inspection of individual regulatory predictions over graphical exploration.

**10.1 Table Structure.** The table displays one row per predicted regulatory edge. The columns are: Source Gene (the transcription factor), Target Gene, Consensus Count (the number of algorithms that predicted this edge), Consensus Score (the mean normalized score across contributing algorithms), and one additional column per selected algorithm showing that algorithm's individual score or rank for the edge. Cells where a given algorithm did not predict the edge are displayed with a dash against a muted background, making coverage gaps immediately visible.

**10.2 Sorting.** Every column is sortable in ascending or descending order. By default, the table is sorted by Consensus Rank ascending (highest-confidence edges first).

**10.3 Filtering and Search.** A text search field filters the table to rows where either the source or target gene name matches the query string. This enables rapid lookup of specific genes.

**10.4 Column Visibility.** When many algorithms are selected, the table may become very wide. A column visibility control allows the user to show or hide individual algorithm columns, focusing the view on the methods of greatest interest.

**10.5 Cross-Linking to Network View.** Clicking any row in the table scrolls and centers the network canvas on the corresponding edge, selects both the source and target nodes, and highlights the edge. This bidirectional navigation between table and graph views supports different analytical workflows.

**10.6 Full-Screen Mode.** The table can be expanded to occupy the full viewport for dense data review, then collapsed back to its default position alongside the network canvas.

# 11. Results Summary and Benchmarking

The results summary provides high-level statistical characterization of the analysis output. It is accessed from the results hub and presents aggregate metrics rather than individual edges.

**11.1 Per-Algorithm Edge Count.** A bar chart shows the number of unique edges predicted by each selected algorithm within the top-N cutoff. This gives the user an immediate sense of how liberal or conservative each method is in its predictions.

**11.2 Method Overlap Visualization.** An overlap visualization (UpSet plot for four or more methods, Venn diagram for two or three) shows the number of edges uniquely predicted by each method and the sizes of all pairwise and higher-order intersections. This directly reveals the degree of agreement or disagreement among methods — a critical interpretive dimension of multi-algorithm analysis.

**11.3 Ground-Truth Benchmarking (Optional).** For expert users who possess a known ground-truth regulatory network (e.g., from a curated database or a synthetic dataset), the platform accepts an optional ground-truth edge list upload. When ground truth is provided, the results summary computes and displays precision-recall curves for each algorithm and for the consensus network, the Area Under the Precision-Recall Curve (AUPRC) for each method, and the AUPRC ratio (the method's AUPRC divided by the AUPRC of a random predictor). These metrics mirror the evaluation methodology of the BEELINE study and allow users to assess method performance on their specific dataset.

# 12. Export and Reproducibility

GRNScope provides comprehensive export functions to ensure that all results can be saved locally, shared with collaborators, and reproduced independently.

**12.1 Network Image Export.** The current state of the network visualization (after all filtering, layout, and selection operations) can be exported as a high-resolution PNG image or as a scalable SVG file. The export captures only the currently visible portion of the graph, including the effects of consensus thresholding and sub-network isolation.

**12.2 Edge List Export.** The complete filtered edge list (reflecting the current consensus threshold and top-N settings) can be exported as a CSV file. The CSV includes columns for Source, Target, Consensus Score, Consensus Rank, Supporting Algorithms (as a comma-separated list), and Algorithm Count. This file is suitable for import into downstream analysis tools such as Cytoscape Desktop, R/igraph, or NetworkX.

**12.3 Individual Algorithm Results Download.** For each completed algorithm, the platform provides a download link to the raw ranked edge list produced by that algorithm's container. These files follow the standardized four-column format (Source, Target, Score, Rank) and are generated directly by the algorithm without any post-processing by the platform.

**12.4 Analysis Metadata Export.** Alongside the edge list, the user may download a metadata file (JSON format) that records all parameters necessary to reproduce the analysis: the job identifier, the original dataset filename and dimensions, the exact preprocessing parameters (gene count, normalization, log-transformation), the list of algorithms executed with their Docker image versions, the consensus threshold and top-N settings at the time of export, and a timestamp.

# 13. Algorithm Extensibility

The platform is designed for continuous expansion of its algorithm library. The extensibility function encompasses both the technical contract that new algorithms must satisfy and the administrative workflow for registering them.

**13.1 Standardized Input/Output Contract.** Every algorithm executes inside an isolated Docker container that conforms to a fixed interface. The container receives a preprocessed gene expression matrix (and optionally a pseudotime file) at a specified filesystem path. It must produce a single ranked edge list CSV at a specified output path and exit with code zero on success. This contract is the only requirement imposed on algorithm implementations; the internal language, libraries, and computational approach are unconstrained.

**13.2 Contributor Workflow.** A contributor wishing to add a new algorithm forks a template repository that provides the Dockerfile skeleton and sample input/output files. The contributor implements their algorithm within this framework, verifies that it produces valid output on the sample data, and submits the Docker image for review. After review and testing by the platform maintainers, the image is deployed to the platform's container registry.

**13.3 Algorithm Registration.** An administrator registers the new algorithm by creating an entry in the algorithm database. The entry specifies the algorithm name, category, capability flags (directed, signed, pseudotime-required), the Docker image tag, the estimated runtime, a textual description, the publication DOI, and the Celery queue assignment (fast or slow). Upon saving, the new algorithm immediately appears in the public algorithm directory and becomes available for selection in the analysis wizard. No application code changes or redeployments are required.

# 14. Landing Page and Public Information

The landing page serves as the entry point for new visitors and provides public-facing information that does not require authentication.

**14.1 Platform Overview.** A hero section communicates the platform's purpose in a single headline and a brief description: GRNScope enables researchers to infer gene regulatory networks from single-cell RNA-seq data using multiple algorithms and to explore consensus results through an interactive interface.

**14.2 Workflow Summary.** A three-step visual graphic illustrates the analysis pipeline: Upload Dataset, Select Algorithms, and Explore Network. This gives first-time visitors an immediate understanding of the user journey.

**14.3 Platform Statistics.** A statistics bar displays live aggregate counts: the number of supported algorithms, the number of supported network types, and the update frequency. These serve as credibility indicators.

**14.4 Feature Highlights.** A grid section highlights the platform's distinguishing capabilities: Docker-based reproducibility (every algorithm runs in a versioned, isolated container), ensemble and individual method views (users can compare algorithms and compute consensus networks), and pseudotime integration (optional trajectory-aware analysis via Slingshot).

**14.5 Recently Added Methods.** A card-based section near the bottom of the page lists the most recently added algorithms, showing each method's name, publication date, and category. This signals that the platform is actively maintained and expanding.

# 15. Responsive Behavior and Accessibility

**15.1 Desktop-First, Tablet-Responsive.** The platform targets desktop browsers as its primary environment, given the visual complexity of network graphs and wide data tables. All pages adapt to tablet-width screens (1024px and below). The sidebar navigation collapses into a drawer on narrow viewports. The results workspace stacks its two panes vertically rather than side-by-side.

**15.2 Table-Only Fallback.** On viewports too narrow to render the network graph effectively, the results workspace offers a "Table Only" view that suppresses the Cytoscape canvas and presents the edge analysis table at full width.

**15.3 Color-Blind Safety.** All visual encodings that use color to convey information (edge type, node type, algorithm category badges) are supplemented with redundant shape, label, or pattern cues. For example, transcription factor nodes are distinguished from target genes by both color and shape (diamond vs. circle), and activating and repressive edges differ in both color and dash pattern.

*This document describes the complete set of user-facing functions provided by GRNScope as of the v2.0 specification. Each function listed here corresponds to one or more implementation tasks in the technical specification and should be verifiable through acceptance testing against the described behavior.*